Linear Interpolation

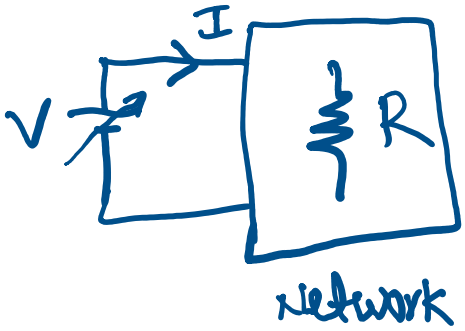
Dr Lokesh Nalla

Digitally signed by Dr Lokesh Nalla
DN: CN=Dr Lokesh Nalla, C=IN, O=ABB, E=lokeshreddy224@gmail.com
Reason: I am the author of this document
Location:
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Interpolation is a process of estimating unknown values that lie between the known data points.

To better understand this concept, consider a look up table populated with data showing the currents through a network at various voltage levels, as shown in figure below.



For *n* voltage samples (V_1, V_2, … V_n), the corresponding currents (I_1, I_2, … I_n) are recorded/measured and tabulated in the look-up table as given below.

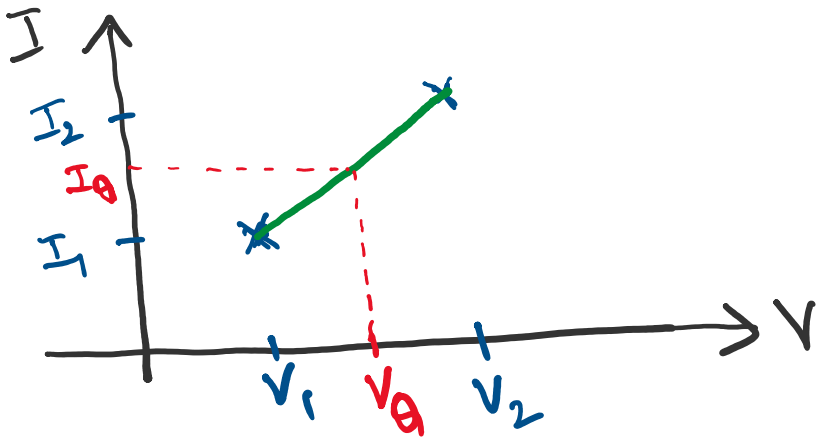
Sample No.	Voltage (V)	Current (A)
1	V_1	I_1
2	V_2	I_2
.	.	.
.	.	.
n	V_n	I_n

To estimate the current (I_Q) at a query voltage point (V_Q) that lies between two known voltages, Interpolation concept is used.

For example, if V_Q is between V_1 and V_2, the updated Look-up table is:

Sample No.	Voltage (V)	Current (A)
1	V_1	I_1
Q	V_Q	?
2	V_2	I_2
.	.	.
.	.	.
n	V_n	I_n

To find the current (I_Q) at the query voltage, plot the voltage-current points on a two-dimensional graph and connect the relevant points with a straight line as shown below.



The equation for the line joining two points allow you to calculate the interpolated current:

$$I_Q = I_1 + \frac{I_2 - I_1}{V_2 - V_1}(V_Q - V_1) \qquad \text{..... (1)}$$

Validation Example:

Assume the network is purely resistive with resistance R = 10 ohms. A subset of Look-up table is as below.

Sample No.	Voltage (V)	Current (A)
1	V_1=10	I_1=1
Q	V_Q=15	?
2	V_2=20	I_2=2

Using the interpolation equation, the current at V_Q = 15 V is estimated as:

$$I_Q = 1 + \frac{2-1}{20-10}(15-10) = 1.5 \text{ A}$$

For verification, calculate the current using Ohm's law:

$$\text{I} = \frac{V}{R} = \frac{15}{10} = 1.5\ A.$$

The interpolated current matches the current calculated using Ohm's law. However, note that interpolation is typically applied to Look-up table data where the resistance value or underlying relationship is unknown.

It is important to note that the underlying relationship should ideally be linear for accurate estimation. If the relationship is non-linear, the sampling interval needs to be significantly reduced to improve accuracy.

For example, consider a look-up table representing a non-linear voltage-power relation instead of a linear voltage-current relationship, as given below:

Sample No.	Voltage (V)	Power (W)
1	V_1=10	P_1=10
Q	V_Q=15	?
2	V_2=20	P_2=40

Using Linear Interpolation, the power at V_Q = 15 V is estimated as:

$$P_Q = P_1 + \frac{P_2 - P_1}{V_2 - V_1}(V_Q - V_1)$$
$$P_Q = 10 + \frac{40-10}{20-10}(15 - 10) = 25\ W$$

However, the actual power at V_Q can be determined using $\mathrm{P} = \frac{V^2}{R} = \frac{15^2}{10} = 22.5\ W$. The deviation between the interpolated and actual values demonstrates that interpolation accuracy depends on the proximity of sample points. To reduce this error, the voltage samples should be taken closer to one another, as shown in the below table:

Sample No.	Voltage (V)	Power (W)
1	V_1=10	P_1=10
Q	V_Q=10.05	?
2	V_2=10.1	P_2=10.201

For $V_Q = 10.05$ V, the interpolated power becomes 10.1005 W and the actual power is 10.10025. This comparison shows that reducing the sampling interval minimizes the percentage error in the estimated value. In summary, finer sampling intervals lead to more accurate results when using linear interpolation for non-linear relationships.

Bilinear Interpolation

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The response/output variable in the look-up table depends on two variables. To estimate the output variable at the query point, Bilinear interpolation is employed.

For example, the below look-up table provides the current values corresponding to various voltages for a resistance R_1 as well as for another resistance R_2.

Sample No.	**Voltage (V)**	**Current at R_1 (A)**	**Current at R_2 (A)**
1	V_1	I_{11}	I_{12}
2	V_2	I_{21}	I_{22}
.	.	.	.
.	.	.	.
n	V_n	I_{n1}	I_{n2}

To estimate the current (I_{QQ}) at a query point (V_Q , R_Q), Bilinear Interpolation concept is used.

For example, if V_Q is between V_1 and V_2, and R_Q is between R_1 and R_2, the updated Look-up table is:

Sample No.	**Voltage (V)**	**Current at R_1 (A)**	**Current at R_Q (A)**	**Current at R_2 (A)**
1	V_1	I_{11}		I_{12}
Q	V_Q	I_{Q1} = ?	I_{QQ} = ?	I_{Q2} = ?
2	V_2	I_{21}		I_{22}
.	.	.		.
.	.	.		.
n	V_n	I_{n1}		I_{n2}

From the above table, it is evident that determining I_{QQ} requires knowing its neighboring currents, I_{Q1} and I_{Q2}.

These currents I_{Q1} and I_{Q2} are estimated using linear interpolation as previously discussed.
To estimate I_{Q1}:
The interpolated current I_{Q1} at (V_Q, R_1) is

$$I_{Q1} = I_{11} + \frac{I_{21}-I_{11}}{V_2-V_1}(V_Q - V_1) \quad \ldots\ldots\ldots\ldots\ldots\ldots (2)$$

To estimate I_{Q2}:
The interpolated current I_{Q2} at (V_Q, R_2) is

$$I_{Q2} = I_{12} + \frac{I_{22}-I_{12}}{V_2-V_1}(V_Q - V_1) \quad \ldots\ldots\ldots\ldots\ldots\ldots (3)$$

To estimate I_{QQ}:
For better understanding, the look-up table is rearranged into a subset of the original table, as shown below:

Resistance (Ω)	**Current (A)**
R_1	I_{Q1}
R_Q	I_{QQ} = ?
R_2	I_{Q2}

Now using linear interpolation equation, the current I_{QQ} at (V_Q, R_Q) is estimated as:

$$I_{QQ} = I_{Q1} + \frac{I_{Q2}-I_{Q1}}{R_2-R_1}(R_Q - R_1) \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (4)$$

The interpolated current I_{QQ} can also be estimated in the other direction, as explained below:
For example, if V_Q lies between V_1 and V_2, and R_Q lies between R_1 and R_2, the updated Look-up table is:

Sample No.	Voltage (V)	Current at R_1 (A)	Current at R_Q (A)	Current at R_2 (A)
1	V_1	I_{11}	I_{1Q} = ?	I_{12}
Q	V_Q		I_{QQ} = ?	
2	V_2	I_{21}	I_{2Q} = ?	I_{22}
.	.	.		.
.	.	.		.
n	V_n	I_{n1}		I_{n2}

From the above table, it is evident that determining I_{QQ} requires knowing its neighboring currents, I_{1Q} and I_{2Q}.

These currents I_{1Q} and I_{2Q} are estimated using linear interpolation as previously discussed.
To estimate I_{1Q}:
A subset of the above look-up table is taken along V_1 to estimate I_{1Q}, as shown below.

Resistance (Ω)	Current (A)
R_1	I_{11}
R_Q	I_{1Q} = ?
R_2	I_{12}

The interpolated current I_{1Q} at (V_1, R_Q) is

$$I_{1Q} = I_{11} + \frac{I_{12}-I_{11}}{R_2-R_1}(R_Q - R_1) \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (5)$$

To estimate I_{2Q}:
The subset of above look-up table is taken along V_2 to estimate I_{2Q}, as shown below.

Resistance (Ω)	Current (A)
R_1	I_{21}
R_Q	I_{2Q} = ?
R_2	I_{22}

The interpolated current I_{2Q} at (V_2, R_Q) is

$$I_{2Q} = I_{21} + \frac{I_{22}-I_{21}}{R_2-R_1}(R_Q - R_1) \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (6)$$

To estimate I_{QQ}:
For better understanding, the look-up table is rearranged into a subset of the original table, as shown below:

Voltage (V)	Current at R_Q (A)
V_1	I_{1Q}
V_Q	I_{QQ} = ?
V_2	I_{2Q}

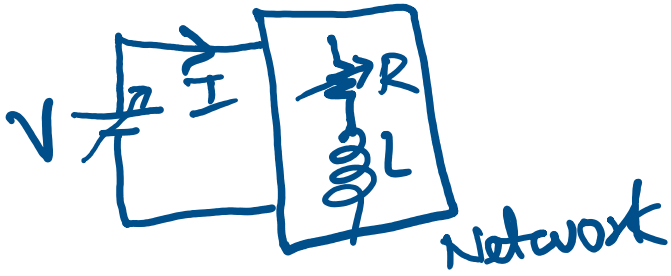
Now using linear interpolation equation, the current I_{QQ} at (V_Q, R_Q) is estimated as:

$$I_{QQ} = I_{1Q} + \frac{I_{2Q} - I_{1Q}}{V_2 - V_1}(V_Q - V_1) \quad \text{..... (7)}$$

Validation Example:

Consider a resistive-inductive network with a fixed reactance of 4 ohms. The magnitude of current values for various voltages and resistances in this network are provided in the following look-up table:

Sample No.	Voltage (V)	Current at R_1 = 3Ω (A)	Current at R_2 = 13Ω (A)
1	$V_1 = 10$	$\|I_{11}\| = 2$	$\|I_{12}\| = 0.7352$
2	$V_2 = 20$	$\|I_{21}\| = 4$	$\|I_{22}\| = 1.4704$



Now estimate the current I_{QQ} at V_Q = 15 V and R_Q = 8 Ω.

The currents I_{Q1} and I_{Q2} from (2) and (3) respectively are obtained as follows:

$$I_{Q1} = 2 + \frac{4-2}{20-10}(15-10) = 3\ A$$

$$I_{Q2} = 0.7352 + \frac{1.4704 - 0.7352}{20-10}(15-10) = 1.1028\ A$$

From (4), the current I_{QQ} is calculated as

$$I_{QQ} = 3 + \frac{1.1028 - 3}{13-3}(8-3) = 2.0514\ A$$

The actual current I_{QQ} is calculated as below:

$$I_{QQ} = \frac{V_Q}{\sqrt{R_Q^2 + 16}} = \frac{15}{8.9442} = 1.677\ A$$

There is a deviation between the estimated current and the actual current, which can be attributed to the large sampling interval. Lets repeat the above example using a smaller sampling interval, with the look-up table provided below:

Sample No.	Voltage (V)	Current at R_1 = 3Ω (A)	Current at R_2 = 13Ω (A)
1	$V_1 = 10$	$\|I_{11}\| = 2$	$\|I_{12}\| = 0.7352$
2	$V_2 = 10.1$	$\|I_{21}\| = 2.02$	$\|I_{22}\| = 0.7425$

Now estimate the current I_{QQ} at V_Q = 10.05 V and R_Q = 8 Ω.

The currents I_{Q1} and I_{Q2} from (2) and (3) respectively are obtained as follows:

$$I_{Q1} = 2 + \frac{2.02-2}{10.1-10}(10.05-10) = 2.01\ A$$

$$I_{Q2} = 0.7352 + \frac{0.7425 - 0.7352}{10.1-10}(10.05-10) = 0.73885\ A$$

From (4), the current I_{QQ} is calculated as

$$I_{QQ} = 2.01 + \frac{0.73885 - 2.01}{13 - 3}(8 - 3) = 1.374425\ A$$

The actual current I_{QQ} is calculated as below:

$$I_{QQ} = \frac{V_Q}{\sqrt{R_Q^2 + 16}} = \frac{10.05}{8.9442} = 1.1236\ A$$

The deviation is still not within an acceptable range, so the sampling interval needs to be reduced further.